

CSCI 131 — Introduction to Programming in Python

Programming Assignment 5

Loops

This assignment exercises the two kinds of loop from this week: **while**, which keeps running its body as long as a condition is **True**, and **for**, which runs its body once for each item in a sequence.

There are no starter files. Open IDLE, choose **File** → **New File** for each program, and save it under the exact name given in the problem. Put your name in a comment on the first line.

Problem 1 — Doubling

Write a program `double.py` which asks for a starting amount and a yearly growth rate, then prints how many years it takes for the amount to at least double. Each year, the amount grows by the rate: an amount of 1000 at a rate of 0.07 becomes 1070 after one year, 1144.9 after two, and so on. Use the prompts `start = float(input('Starting amount: '))` and `rate = float(input('Growth rate: '))`, in that order. Your program should produce these answers:

Starting amount	Rate	Output
1000	0.07	Years to double: 11
500	0.10	Years to double: 8
2000	0.03	Years to double: 24

Problem 2 — Factorial

Write a program `factorial.py` which asks for a whole number n and prints n factorial, which is written $n!$ and is the product of every whole number from 1 up to n . For example, $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$. By definition, $0! = 1$. Use the prompt `n = int(input('n: '))`. You may assume the user types a whole number that is zero or more.

Your program should produce these answers:

n	Output
5	120
10	3628800
0	1

Submission

Submit both source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.

If you are not finished when the period ends, submit what you have. You may complete it on your own and resubmit by Friday, October 9 at 11:59 pm for up to 50% of the remaining credit.